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Stripe (Payment Processing)

Delivery 1 — Requirements & Workload Specification

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1 System Description

1.1 Business Problem

In the real world, as technology has evolved, making online payments through physical and virtual stores has become part of everyday life. Businesses need reliable payment processing systems that allow them to manage transactions from a centralized platform, track sales, and access accurate financial information. At the same time, buyers and sellers need to trust that their payment information and money are protected against unauthorized transactions and fraud.

For this reason, there is a need for platforms capable of processing payments securely, maintaining accurate transaction records, and detecting potentially fraudulent activity through appropriate computational strategies. The primary beneficiaries are sellers and businesses, which can process payments and manage their transactions through a centralized system, while buyers benefit from secure and reliable payment experiences.

However, implementing such a platform at a large scale presents significant operational challenges. The system must be capable of processing a very high volume of transactions while maintaining low response times, strict consistency for financial records, and zero data loss. It must also protect sensitive information, detect fraud in real time, coordinate with external payment networks, and remain highly available. Therefore, scalability, security, reliability, and consistency are critical considerations when designing and operating the system.

1.2 Actors

The following actors represent the human users and external systems that interact with the payment processing system.

Human Actors

- **Administrator**
  - **Role:** Manages platform configuration, payment settings, user access, security rules, and system operations.
  - **Interaction Frequency:** Frequent, mainly for monitoring, configuration, user management, and operational activities.
  - **Expectations:** Secure access, reliable transaction information, effective configuration tools, real-time monitoring, and high system availability.
- **Auditor**
  - **Role:** Reviews transaction records, audit trails, financial operations, and reconciliation information to verify traceability and compliance.
  - **Interaction Frequency:** Periodic, mainly during audits, investigations, reconciliation processes, and compliance reviews.
  - **Expectations:** Complete and immutable audit trails, accurate financial records, traceable transactions, consistent data, and secure access to historical information.

- **Buyer**
  - **Role:** Initiates payments for products or services and receives confirmation of successful, failed, or pending transactions.
  - **Interaction Frequency:** Occasional or frequent, depending on the number of purchases made.
  - **Expectations:** Fast and secure payments, accurate transaction status, protection of personal and payment information, and reliable payment confirmations.
- **Seller**
  - **Role:** Receives payments from Buyers and manages payment-related information, transaction statuses, refunds, balances, and settlements.
  - **Interaction Frequency:** Frequent, especially for monitoring transactions, payment confirmations, refunds, and settlements.
  - **Expectations:** Reliable payment processing, accurate transaction records, timely settlement information, secure access, and reliable notifications for relevant payment events.

External System Actors

- **Banking and Payment Networks**
  - **Role:** Processes payment authorization and transaction requests received from the payment processing system and returns the corresponding transaction results.
  - **Interaction Frequency:** Continuous, whenever payment transactions are submitted for authorization or processing.
  - **Expectations:** Valid transaction requests, secure communication, reliable responses, and accurate transaction processing.
- **External Notification Providers**
  - **Role:** Delivers payment-related notifications generated by the payment processing platform through external communication channels such as email, SMS, or push notifications.
  - **Interaction Frequency:** Frequent, whenever a payment event requires an external notification.
  - **Expectations:** Accurate notification content, reliable message delivery, and timely responses regarding delivery status.

Based on the actors and system interactions previously identified, the following diagram presents a systemic view of the proposed payment processing platform. It illustrates the main actors and external systems interacting with the system, as well as the primary subsystems that collaborate internally to provide the required payment processing services.

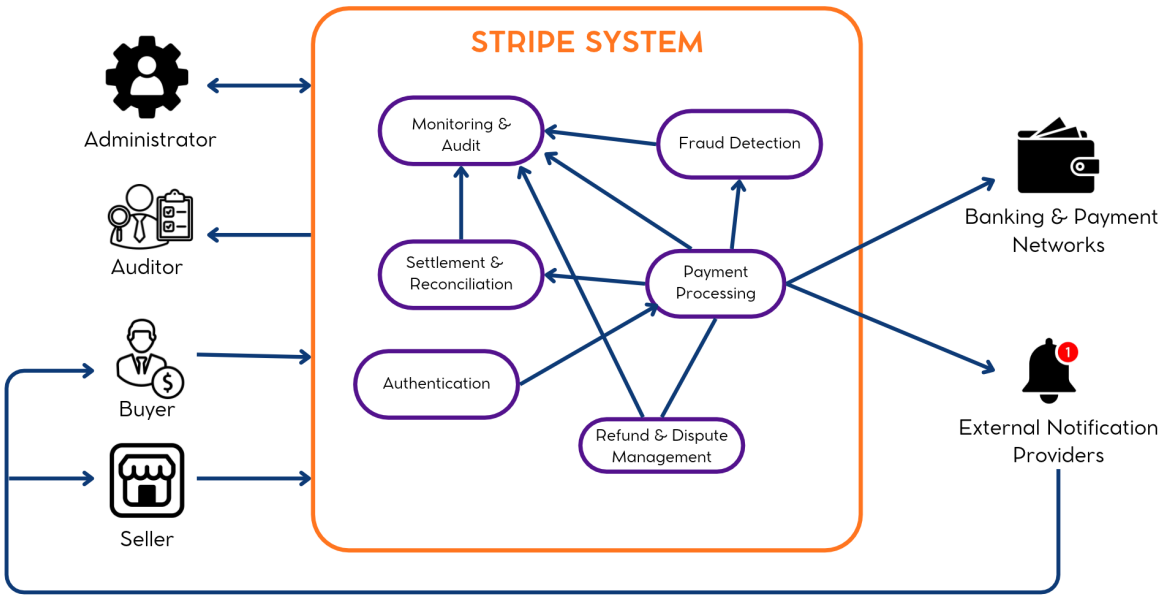


Figure 1: Stripe System Diagram

1.3 Main Use Cases

The following use cases describe the primary interactions between the actors and the payment processing system. Each use case defines the actors involved, preconditions, main flow, alternative flows, and postconditions to specify the expected behavior of the system in different transaction scenarios.

Title: UC-01: Make a Payment		
Actor(s): Buyer, Seller, Payment Network	**Priority:** High	**Estimate:** 8 Story Points

User Story:
As a Buyer,
I want to make a payment for a product or service,
so that the transaction can be processed securely and the Seller can receive the payment.

Preconditions:
The Buyer has selected a product or service to purchase. The Seller is registered in the system and the required payment information is available.

Main Flow:
1. The Buyer initiates the payment.
2. The Buyer provides the required payment information.
3. The Buyer submits the payment request.
4. The system validates and processes the payment.
5. The system displays the payment result to the Buyer.
6. The Seller receives the corresponding transaction status.

Alternative Flows:
2a. The payment information is invalid. The system requests the Buyer to correct the information.
3a. The Payment Network declines the transaction. The system informs the Buyer that the payment was not completed.
4a. The transaction is identified as potentially fraudulent. The system places the transaction under review or rejects it.
4b. The transaction cannot be completed immediately. The system records it as pending and provides the corresponding status.

Postconditions:
The transaction is recorded with its resulting status, and the Buyer and Seller can access the corresponding payment information.

Acceptance Criteria:
Given that the Buyer has valid payment information,
When the Buyer submits the payment,
Then the system shall process the transaction and provide its resulting status.

Title: UC-02: Check Payment Status		
Actor(s): Buyer, Seller	**Priority:** High	**Estimate:** 3 Story Points
User Story: As a Buyer or Seller, I want to check the status of a payment, so that I can know whether the transaction was completed, declined, or remains pending.		
Preconditions: A payment transaction exists in the system and the actor has permission to access its information.		
Main Flow: 1. The Buyer or Seller requests the status of a transaction. 2. The system identifies the requested transaction. 3. The system retrieves the current transaction status. 4. The system displays the payment status and relevant information.		
Alternative Flows: **1a.** The actor does not have permission to access the transaction. The system denies access. **2a.** The requested transaction does not exist. The system informs the actor that the transaction could not be found. **3a.** The transaction is still being processed. The system displays a pending status.		
Postconditions: The actor receives the current status and relevant information of the requested transaction.		
Acceptance Criteria: Given that a valid transaction exists, When the Buyer or Seller requests its status, Then the system shall display the current status of the transaction.		

Title: UC-03: Handle Suspicious Payment		
Actor(s): Buyer, Administrator, Fraud Detection	**Priority:** High	**Estimate:** 8 Story Points

User Story:
As an Administrator,
I want the system to identify and handle potentially fraudulent payments,
so that suspicious transactions can be reviewed or prevented.

Preconditions:
A payment transaction has been submitted and is subject to fraud analysis.

Main Flow:
1. The system sends the transaction for fraud analysis.
2. The Fraud Detection System evaluates the transaction.
3. The system receives the resulting risk information.
4. The system determines the corresponding transaction status.
5. The Administrator can review transactions identified as suspicious.

Alternative Flows:
2a. The Fraud Detection System identifies a high-risk transaction. The system prevents or places the transaction under review.
2b. The Fraud Detection System does not identify suspicious activity. The transaction continues through the normal payment process.
3a. The fraud analysis service is unavailable. The system handles the transaction according to the defined security policy.

Postconditions:
The transaction has a recorded fraud-related result and, when necessary, is available for Administrator review.

Acceptance Criteria:
Given that a payment is submitted for processing,
When the transaction is analyzed for fraud,
Then the system shall record the resulting risk status and apply the corresponding transaction handling.

Title: UC-04: Request a Refund		
Actor(s): Seller, Buyer, Payment Network	**Priority:** High	**Estimate:** 5 Story Points

User Story:
As a Seller,
I want to request a refund for an eligible transaction,
so that the corresponding amount can be returned to the Buyer.

Preconditions:
A completed transaction exists and is eligible for a refund. The Seller has permission to request the refund.

Main Flow:
1. The Seller selects an eligible transaction.
2. The Seller requests a refund.
3. The system validates the refund request.
4. The system processes the refund through the corresponding payment channel.
5. The system records the refund result.
6. The Buyer and Seller receive the corresponding refund status.

Alternative Flows:
1a. The transaction is not eligible for a refund. The system rejects the request and informs the Seller.
3a. The requested refund amount is invalid. The system requests correction of the refund information.
4a. The Payment Network rejects the refund. The system records the failed result and informs the Seller.

Postconditions:
The refund request has a recorded status and the corresponding parties can access the result.

Acceptance Criteria:
Given that an eligible transaction exists,
When the Seller requests a refund,
Then the system shall process the request and record its resulting status.

Title: UC-05: Receive Payment Notification		
Actor(s): Seller, Seller's System	**Priority:** High	**Estimate:** 5 Story Points
User Story: As a Seller, I want to receive notifications about payment events so that I can keep my records and processes updated.		
Preconditions: 1. A payment-related event has occurred. 2. The Seller has an active notification mechanism.		
Main Flow: 1. A payment-related event occurs. 2. The system generates the corresponding notification. 3. The system sends the notification to the Seller or Seller's System. 4. The Seller's System receives the event. 5. The system records the notification status.		
Alternative Flows: **1a.** If the notification cannot be delivered, the system retries the delivery according to the configured mechanism. **2a.** If delivery continues to fail, the system records the failed notification. **3a.** If the Seller's System is temporarily unavailable, the notification remains available for subsequent delivery.		
Postconditions: The payment event has been delivered or recorded as undelivered for subsequent handling.		
Acceptance Criteria: Given that a payment event occurs, When the system generates the corresponding notification, Then the Seller or Seller's System shall receive the relevant transaction information.		

Title: UC-06: View Balance and Settlement Information		
Actor(s): Seller, Settlement System	**Priority:** High	**Estimate:** 5 Story Points
User Story: As a Seller, I want to view my balance and settlement information so that I can track the funds generated by my transactions.		
Preconditions: 1. The Seller has an active account. 2. The Seller has recorded transactions.		
Main Flow: 1. The Seller accesses the balance and settlement section. 2. The system retrieves the Seller's financial information. 3. The system displays the available balance and relevant settlement information. 4. The Seller reviews the information.		
Alternative Flows: **1a.** If there are no transactions available, the system displays the corresponding information. **2a.** If settlement information is temporarily unavailable, the system informs the Seller and allows the information to be consulted later.		
Postconditions: The Seller can view the available balance and settlement information associated with the account.		
Acceptance Criteria: Given that the Seller has recorded transactions, When the Seller accesses the balance information, Then the system shall display the corresponding financial information accurately.		

Title: UC-07: Manage Payment Dispute		
Actor(s): Seller, Administrator	**Priority:** Medium	**Estimate:** 8 Story Points

User Story:
As a Seller, I want to review and respond to payment disputes so that disputed transactions can be properly handled.

Preconditions:
1. A payment dispute exists.
2. The Seller or Administrator is authorized to access the dispute.

Main Flow:
1. The Seller accesses the list of payment disputes.
2. The system displays the available dispute information.
3. The Seller reviews the disputed transaction.
4. The Seller submits the required response or information.
5. The system records the response.
6. The Administrator can review the dispute and its history.

Alternative Flows:
1a. If the Seller does not respond within the applicable period, the system records the missed response.
2a. If the information submitted is incomplete, the system requests the missing information.
3a. If the dispute requires administrative intervention, the Administrator reviews the case.

Postconditions:
The dispute and its corresponding actions are recorded for further processing and traceability.

Acceptance Criteria:
Given that a payment dispute exists,
When the Seller submits a response,
Then the system shall record the response and make it available for authorized review.

Title: UC-08: Monitor Payment Operations		
Actor(s): Administrator	**Priority:** High	**Estimate:** 8 Story Points

User Story:
As an Administrator, I want to monitor payment operations so that I can identify and respond to operational issues.

Preconditions:
1. The Administrator is authenticated and authorized.
2. The system is collecting operational information.

Main Flow:
1. The Administrator accesses the monitoring interface.
2. The system displays relevant payment and operational information.
3. The Administrator reviews the current system status.
4. The system identifies relevant operational events or failures.
5. The Administrator takes the appropriate administrative action.

Alternative Flows:
1a. If an external service becomes unavailable, the system records the event and informs the Administrator.
2a. If a payment processing component becomes unavailable, the system reports the incident.
3a. If an alternative processing mechanism is available, the system may continue processing according to the configured operational rules.

Postconditions:
Operational events are recorded and the Administrator has access to the information required for monitoring and response.

Acceptance Criteria:
Given that the system is processing transactions,
When an operational event occurs,
Then the system shall record and make the relevant information available to the Administrator.

Title: UC-09: Verify Seller		
Actor(s): Administrator, Seller	**Priority:** Medium	**Estimate:** 5 Story Points
User Story: As an Administrator, I want to review and verify Seller information so that only authorized Sellers can use the payment platform.		
Preconditions: 1. A Seller account has been registered. 2. The Administrator has authorization to review Seller information.		
Main Flow: 1. The Seller provides the required information. 2. The Administrator accesses the Seller's verification request. 3. The Administrator reviews the submitted information. 4. The Administrator approves or rejects the verification request. 5. The system records the verification result. 6. The Seller receives the corresponding status.		
Alternative Flows: **1a.** If the information is incomplete, the Administrator requests additional information. **2a.** If the information does not satisfy the applicable requirements, the Administrator rejects the verification request. **3a.** If additional review is required, the verification remains pending.		
Postconditions: The Seller's verification status is recorded and available to authorized users.		
Acceptance Criteria: Given that a Seller has submitted the required information, When the Administrator completes the review, Then the system shall record and communicate the resulting verification status.		

Title: UC-10: Reconcile Transactions		
Actor(s): Auditor, Settlement System	**Priority:** High	**Estimate:** 8 Story Points
User Story: As an Auditor, I want to reconcile transaction records so that financial information can be verified and discrepancies can be identified.		
Preconditions: 1. Transaction records are available. 2. Settlement information is available for the period being reviewed. 3. The Auditor is authorized to access the records.		
Main Flow: 1. The Auditor selects the transactions or period to review. 2. The system retrieves the corresponding transaction records. 3. The system retrieves the available settlement information. 4. The system compares the relevant financial records. 5. The system displays the reconciliation results. 6. The Auditor reviews the identified information.		
Alternative Flows: **1a.** If a transaction does not have corresponding settlement information, the system identifies it as requiring review. **2a.** If a discrepancy is identified, the system records the discrepancy for further investigation. **3a.** If the required information is incomplete, the system indicates the missing information.		
Postconditions: The reconciliation results and any identified discrepancies are recorded and available for authorized review.		
Acceptance Criteria: Given that transaction and settlement records are available, When the Auditor performs a reconciliation, Then the system shall compare the records and identify any discrepancies.		

1.4 System Boundaries

This section formally defines the *System Boundary* of the "Stripe (Payment Processing)" project, based on the reference architecture diagram. This boundary explicitly delineates the computational components managed under the system's primary engineering control from external dependencies. It also serves as a critical mapping point for integration contracts, data flows, and regulatory security responsibilities (NFR-11, PCI DSS scope).

1.4.1 Internal Scope (In-Boundary Modules)

The internal processing domain is encapsulated by a single high-security boundary. The internal architecture consists of six interdependent modules responsible for fulfilling the primary functional

requirements and use cases:

1. **Authentication:** Gateway responsible for establishing user identity, validating tokens, and enforcing the secure Role-Based Access Control (RBAC) defined for both human and system actors.
2. **Fraud Detection:** Real-time evaluation engine applying computational strategies to assess transaction risk profiles. It provides deterministic outputs (< 100 ms as per NFR-05) before authorizing payment execution (UC-03).
3. **Payment Processing:** The central workflow engine coordinating UC-01 (Make a Payment). It orchestrates data exchange between Fraud Detection and external banking platforms to authorize and capture funds.
4. **Settlement & Reconciliation:** Business logic for managing account balances, processing fund ledger entries, and validating consistency against external third-party statements (Supports UC-06, UC-10).
5. **Monitoring & Audit:** The central intelligence and compliance hub, recording immutable financial logs for auditability (UC-10, NFR-07) and continuously evaluating operational health (UC-08, NFR-06).
6. **Refund & Dispute Management:** Handles exceptional transactional lifecycles and state machines required for chargebacks, claims, and refunds (UC-04, UC-07).

The 100 ms bound of Fraud Detection corresponds to the 95th percentile defined in NFR-05 and excludes external network latency. Fraud Detection and Settlement & Reconciliation are internal modules of the platform: the "Fraud Detection System" and "Settlement System" named as actors in UC-03, UC-06, and UC-10 refer to these modules and not to external parties (see Table 2).

Table 1 allocates every use case and functional requirement to the module that fulfils it, including those not named in the module descriptions above (seller verification and notification delivery).

Table 1: Allocation of Use Cases and Functional Requirements to Internal Modules

Module	**Use Cases**	**Functional Req.**	**Responsibility**
Authentication	All UCs (access control); UC-09	FR-11, FR-12	Identity, RBAC, and MFA for every actor; registration data and verification status of Sellers submitted by Sellers and reviewed by Administrators.
Fraud Detection	UC-03	FR-04	Risk evaluation of each transaction and recording of the risk result; suspicious transactions are exposed for Administrator review.
Payment Processing	UC-01, UC-02, UC-03	FR-01, FR-02, FR-03	Validation, processing, and status of payments; applies the fraud decision and, if Fraud Detection is unavailable, the defined security policy (UC-03, alternative 3a); tokenization of card data.
Settlement & Reconciliation	UC-06, UC-10	FR-07, FR-13, FR-14	Balances, ledger entries, and settlement information; comparison against external statements and flagging of discrepancies.
Monitoring & Audit	UC-05, UC-08, UC-10	FR-06, FR-10	Operational monitoring, immutable audit records, and generation, delivery, and status tracking of event notifications to Sellers' Systems and Notification Providers.
Refund & Dispute Management	UC-04, UC-07	FR-05, FR-08, FR-09	Refund processing and dispute lifecycle, including exchanges with the Banking & Payment Networks.

1.4.2 External Scope (Out-of-Boundary Entities)

The seven entities operating externally to the Stripe System Boundary communicate strictly through dedicated interfaces. In this section, (H) denotes a human actor and (S) a system actor.

Human Actors (Users) These actors interact via web or mobile client interfaces built against public and administrative APIs:

- **Administrator (H):** Accesses system configuration, verification routines, and operational controls.
- **Auditor (H):** Accesses historical, immutable audit trails and regulatory ledgers.
- **Buyer (H):** Submits payment data and receives real-time transaction states.
- **Seller (H):** Manages merchant balance, dispute resolutions, and settlement reports.

External System Actors (Downstream Dependencies) Third-party platforms and protocols maintained by external organizations:

- **Banking & Payment Networks (S):** External networks (e.g., Visa, Mastercard, ACH) executing the settlement of funds and official authorization decisions.
- **External Notification Providers (S):** Third-party services (e.g., Twilio, Mailgun, FCM) delivering asynchronous multi-channel event notifications (UC-05).
- **Seller's System (S):** Third-party e-commerce or ERP systems registered to receive cryptographically signed webhook notifications (UC-05).

Table 2 relates the actors named in the use cases and in the actor list to the entities and modules of this section.

Table 2: Mapping of Use Case Actors to Boundary Entities and Modules

Actor in Use Cases	**Boundary Element**	**Note**
Administrator, Auditor, Buyer, Seller	External human actors (H)	Same actors defined in the actor list.
Payment Network (UC-01, UC-04)	Banking & Payment Networks (S)	Same entity as "Banking and Payment Networks" in the actor list.
Fraud Detection System (UC-03)	Internal module: Fraud Detection	Not an external party.
Settlement System (UC-06, UC-10)	Internal module: Settlement & Reconciliation	Not an external party; it validates against statements provided by the Banking & Payment Networks.
Seller's System (UC-05)	Seller's System (S)	External system actor; it also receives events.
External Notification Providers	External Notification Providers (S)	External system actor defined in the actor list.

1.4.3 System Boundary Interfaces and Integration Contracts

Table 3 summarizes the contractual obligations, target touchpoints, and security protocols enforced at each boundary edge. Latency figures follow NFR-05 and NFR-10 and exclude the latency introduced by external payment networks.

Table 3: System Boundary Interfaces and Integration Contracts

External Entity	Boundary Touchpoint	Interaction Description	Security / Latency Contract
Administrator (H)	Authentication, Monitoring & Audit, Fraud Detection, Refund & Dispute Management	Monitors operational health, verifies sellers, configures rules, and reviews suspicious transactions and disputes.	HTTPS (TLS 1.3); MFA mandatory; RBAC with least privilege.
Auditor (H)	Authentication, Monitoring & Audit, Settlement & Reconciliation	Requests immutable audit logs and reconciliation reports.	HTTPS (TLS 1.3); MFA; strict read-only RBAC.
Buyer (H)	Authentication, Payment Processing	Initiates transactions, submits card data to the tokenization endpoint, and queries transaction status.	HTTPS (TLS 1.3); synchronous API latency < 500 ms (p95) and < 2 s (p99) (NFR-05).
Seller (H)	Authentication, Payment Processing, Settlement & Reconciliation, Refund & Dispute Management	Submits verification data, reviews merchant ledger and settlements, handles disputes, and triggers refunds.	HTTPS (TLS 1.3); MFA for sensitive actions (e.g., refunds); RBAC scoped to the seller's own data; same latency as Buyer.
Banking Networks (S)	Payment Processing, Refund & Dispute Management, Settlement & Reconciliation	Gateway link to authorize, capture, or reverse fund movements, process refunds and chargeback events, and provide settlement statements.	Dedicated private connectivity (VPN or private link) with mutual TLS; PCI-compliant gateway; idempotency keys on every request.
Notification Providers (S)	Monitoring & Audit	Asynchronous delivery of critical transactional events to Buyers and Sellers (UC-01, UC-04, UC-05).	Encrypted REST API (HTTPS); at-least-once delivery queue with retries and delivery-status tracking; no PAN in payloads.
Seller's System (S)	Monitoring & Audit	Subscribes to event-driven transactional updates.	HTTPS webhooks with HMAC SHA-256 signatures and timestamp (anti-replay); at-least-once delivery with retries and unique event IDs for idempotent handling.

1.4.4 Trust Boundaries (PCI DSS v4.0.1)

In compliance with **NFR-11** and the **PCI DSS v4.0.1** standard, strict trust boundaries isolate the Cardholder Data Environment (CDE) and keep its scope as small as possible. Within the high-security boundary, the platform is divided into a public zone (edge services and the Authentication module), an application zone, and the CDE:

- **PAN Ingestion & Tokenization:** Public endpoints accessed by Buyers and Sellers never

expose the cleartext Primary Account Number (PAN) after ingestion. Card data is tokenized at the entry point; the PAN exists only inside the CDE, where it is stored encrypted and unreadable (NFR-11), and every other module works exclusively with tokens.

- **CDE Scope:** The CDE is limited to the tokenization service, the token vault, and the Payment Processing connection to the Banking & Payment Networks. Fraud Detection, Settlement & Reconciliation, Refund & Dispute Management, and Monitoring & Audit operate outside the CDE on tokens and derived data. Logs, events, and notification payloads must never contain the PAN.
- **Network Isolation:** The Banking and Payment Network connection uses dedicated private connectivity with mutual TLS, logically segregated from external notification channels and public web infrastructure. Network segmentation applies default-deny rules between the public zone, the application zone, and the CDE.
- **Key Management:** Encryption keys are managed in a KMS/HSM with rotation, separation of duties, and restricted access, covering data in transit and at rest.
- **Strong Authentication:** MFA is required for administrative access and for any access to the CDE.

1.4.5 Deployment Context and Exclusions

Cloud context. The internal modules are deployed on a cloud provider across multiple regions and availability zones to meet NFR-06, NFR-07, and NFR-10. The cloud provider follows a shared-responsibility model: it secures the underlying infrastructure, while the platform remains responsible for configuration, identity, data protection, and application security. Managed edge services (CDN, WAF, DNS) in front of the Authentication module are treated as part of the boundary.

Out of scope. Card issuing, the sellers' product catalogs and checkout logic, the internal operations of the Banking & Payment Networks, and the web and mobile client applications are outside the system; only the API contracts they consume are defined here.

2 Functional Requirements

Functional Requirements

The functional requirements are organized by functional area or service domain. Each requirement is assigned a unique identifier and traced to the corresponding use case.

Payment Processing

ID	Functional Requirement	Related UC
FR-01	The system shall validate payment information before processing a payment request.	UC-01
FR-02	The system shall process payment requests and maintain their corresponding transaction status.	UC-01, UC-02
FR-03	The system shall provide the current status of a payment transaction to an authorized Buyer or Seller.	UC-02

Fraud Detection

ID	Functional Requirement	Related UC
FR-04	The system shall analyze payment transactions for potentially fraudulent activity before completing the transaction.	UC-03

Refund Processing

ID	Functional Requirement	Related UC
FR-05	The system shall allow an authorized Seller to submit and process refund requests for eligible completed payments.	UC-04

Payment Notifications

ID	Functional Requirement	Related UC
FR-06	The system shall generate and deliver notifications for relevant payment events to the Seller or Seller's System.	UC-05

Balance and Settlement

ID	Functional Requirement	Related UC
FR-07	The system shall display to an authorized Seller the available balance and the settlement information associated with their account.	UC-06

Dispute Management

ID	Functional Requirement	Related UC
FR-08	The system shall allow an authorized Seller to view the list of payment disputes and the details of each disputed transaction.	UC-07
FR-09	The system shall allow an authorized Seller to submit a response and supporting information for a dispute, and shall request missing information when the submission is incomplete and record the response.	UC-07

Operational Monitoring

ID	Functional Requirement	Related UC
FR-10	The system shall provide an authenticated and authorized Administrator with an interface displaying payment and operational information, including the current system status.	UC-08

Seller Verification

ID	Functional Requirement	Related UC
FR-11	The system shall allow a registered Seller to submit the information required for verification.	UC-09
FR-12	The system shall allow an authorized Administrator to review a verification request and approve it, reject it, request additional information, or keep it pending, record the result and communicate it to the seller.	UC-09

Transaction Reconciliation

ID	Functional Requirement	Related UC
FR-13	The system shall allow an authorized Auditor to select the transactions or period to be reconciled, compare the selected transaction records with the corresponding settlement information, and display the reconciliation results.	UC-10
FR-14	The system shall identify and flag transactions with missing settlement information or discrepancies, record them for further investigation, and indicate the missing information required to complete the reconciliation.	UC-10

3 Non-Functional Requirements

The non-functional requirements define measurable capacity, performance, availability, reliability, data, security, and geographic targets for the payment processing platform. The request-rate values are based on the scale defined for the project, while values that are not explicitly specified are treated as engineering targets.

Capacity Requirements

ID	Requirement	Verification	Related UC
NFR-01	The system shall support at least 5,000,000 registered platform accounts, with a capacity target of 50,000 concurrent active accounts under normal conditions and 200,000 concurrent active accounts during peak conditions.	Load testing	All UCs
NFR-02	The system shall support at least 20,800,000 incoming API requests per hour under normal operating conditions.	Load testing	All UCs
NFR-03	The system shall support at least 98,600,000 incoming API requests per hour during peak operating periods.	Stress testing	All UCs
NFR-04	The system shall sustain peak operating conditions for at least 4 consecutive hours without exceeding the defined response-time, availability, or data-integrity limits.	Endurance testing	All UCs

Performance Requirements

ID	Requirement	Verification	Related UC
NFR-05	The system shall process synchronous API requests in less than 500 ms at the 95th percentile and less than 2 s at the 99th percentile. For payment authorization requests, the system shall produce a fraud-risk decision in less than 100 ms at the 95th percentile, excluding latency introduced by external payment networks.	Performance testing	UC-01, UC-02, UC-03

Availability & Reliability Requirements

ID	Requirement	Verification	Related UC
NFR-06	The system shall maintain at least 99.99% monthly availability for the public payment-processing API, corresponding to a maximum of approximately 4.32 minutes of unplanned downtime during a 30-day month.	Availability monitoring	TBD
NFR-07	The system shall maintain a Recovery Point Objective (RPO) of zero for all acknowledged financial transactions and ledger entries in the event of a single availability-zone failure.	Failure and recovery testing, Availability monitoring	All UCs

Data Requirements

ID	Requirement	Verification	Related UC
NFR-08	The system shall support at least 1 TB of logical application data at the initial production deployment, including payment records, ledger entries, audit records, customer and seller data, and idempotency records.	Capacity testing	UC-01, UC-04, UC-06, UC-10
NFR-09	The system shall support an increase of at least 3 TB of logical application data per year without requiring a redesign of the logical data model.	Capacity and growth testing	TBD

Geographic Scope Requirements

ID	Requirement	Verification	Related UC
NFR-10	The system shall provide payment-processing services globally through geographically distributed infrastructure, with a target application-layer response time below 500 ms at the 95th percentile for requests processed without third-party payment-provider latency.	Distributed performance testing	TBD

Security & Compliance Requirements

ID	Requirement	Verification	Related UC
NFR-11	The system shall encrypt sensitive payment and financial data at rest and in transit, render stored Primary Account Numbers (PAN) unreadable, and implement and document the applicable PCI DSS v4.0.1 controls within the project scope.	Security and compliance testing	TBD

4 Workload Model

Table 4: Stripe Workload Model

ID	Parameter	Value
WL-01	Average requests/hour	$\approx$ 20.83 million req/h
WL-02	Peak requests/hour	$\approx$ 98.62 million req/h
WL-03	Peak duration	4-day(96 hours) peak activity period;
WL-04	Read/Write ratio	35%/65%
WL-05	Availability target (SLA)	99.999% reliability target
WL-06	Response time target	P95 target of $\leq 200ms$
WL-07	Estimated data volume at launch	150 TB raw / 195 TB logical / 585 TB physical
WL-08	Annual data growth	34% anual
WL-09	Concurrent users	$\approx$ 2740
WL-10	Geographic scope	Global; Operations across 185 countries;

4.1 Workload Model Justification

4.1.1 WL-01: Average Requests per Hour

Stripe reports processing more than 500 million API requests per day [1]. Assuming a 24-hour day, the corresponding average hourly request rate is:

$$R_{\text{avg}} > \frac{500 \times 10^6}{24} \approx 20.83 \times 10^6 \text{ requests/hour.} \quad (1)$$

This value represents a daily-average rate converted to an hourly equivalent, rather than a directly reported hourly measurement. Actual traffic may vary throughout the day.

4.1.2 WL-02: Peak Requests per Hour

During Black Friday and Cyber Monday (BFCM) 2023, Stripe reported a peak API request rate of 27,395 requests per second [2]. Converting this instantaneous peak rate to an hourly equivalent gives:

$$R_{\text{peak,hour}} = 27{,}395 \times 3{,}600 = 98{,}622{,}000 \text{ requests/hour.} \quad (2)$$

This result represents the hourly equivalent of the reported peak rate. It does not imply that Stripe sustained this rate throughout an entire hour.

4.1.3 WL-03: Peak Duration

Stripe reported that Black Friday through Cyber Monday (BFCM) 2023 was its largest four-day period to date, during which it processed more than 300 million transactions and recorded a peak API request rate of 27,395 requests per second [1,2]. Therefore, four days (96 hours) can be used to describe the observed high-activity period. However, Stripe does not disclose how long the maximum request rate of 27,395 requests per second was sustained.

4.1.4 WL-04: Read/Write Ratio

Stripe does not publicly disclose the global read/write ratio of its production workload. We propose a workload distribution of 35% read operations and 65% write operations for the transactional and ledger layer. This propose is based on the load test published by Fragment [3], which reported 6,769,966 read requests and 12,852,643 write requests out of 19,622,609 total requests during a 15-minute test. These figures correspond to approximately 34.5% reads and 65.5% writes, respectively. Accordingly, the proposed 35/65 ratio is a modeling assumption rather than an observed Stripe metric.

4.1.5 WL-05: Availability Target

Stripe states that it targets five nines of reliability on an ongoing basis and reports API availability greater than 99.999% during BFCM 2023 [1, 2]. Accordingly, 99.999% is adopted as the availability target for this workload model. This is a documented reliability target and an observed result for the reported period; it should not be interpreted as a contractual SLA for every Stripe product or endpoint.

For a 365-day year, 99.999% availability corresponds to an allowance of approximately 5.26 minutes of unavailability, assuming a continuous service and that the percentage applies to the full year.

4.1.6 WL-06: Response Time Target

Stripe does not publicly disclose absolute P95 and P99 latency measurements for its global API workload. However, its engineering publications provide relevant references for establishing latency objectives. Stripe reports that its Radar fraud detection system makes fraud decisions in less than 100 milliseconds [4]. This figure applies specifically to fraud decisioning and should not be interpreted as the end-to-end response time of the entire payment API.

Additionally, Stripe has reported that a redesigned architecture for its core API calls improved 99th-percentile latency by more than 50% for pilot customers, although the publication does not provide an absolute latency value in milliseconds [5].

Based on these references, the proposed workload model establishes a P95 response-time target of 200 ms for synchronous API requests.

4.1.7 WL-07: Estimated Data Volume at Launch

Stripe does not publicly disclose the initial data volume of its platform at launch. Stripe reports that its Ledger system processes approximately five billion events per day [6]. This figure provides a reference for the scale of a mature financial event processing system, although it does not represent five billion independent payments. For capacity planning, we assume an average event size of 1 KB. This is a design assumption informed by the variable record sizes found in transactional database benchmarks such as TPC-C [7, 8], rather than a measured Stripe record size.

Under these assumptions, the raw data volume is approximately 5 TB per day. For an initial 30-day data window, the estimated raw volume is 150 TB. We then apply a 30% planning allowance for indexes and storage metadata, resulting in approximately 195 TB of logical storage. Database indexes introduce additional storage overhead, although the actual amount depends on the schema and indexing strategy [9].

Finally, assuming three complete copies of the stored data for the purpose of this capacity-planning scenario, the estimated physical storage requirement is 585 TB.

4.1.8 WL-08: Annual Data Growth

Stripe does not publicly disclose a precise annual growth rate for the storage volume of its production platform. However, its public reports provide indicators that can support a transparent planning assumption. In 2025, the total payment volume processed by businesses using Stripe increased by 34% compared with 2024, following a reported increase of 38% in 2024 [10, 11]. Although payment volume is not equivalent to stored data volume, it provides an indirect indicator of the growth in platform activity and is therefore used as the basis for the proposed scenario.

Additional evidence highlights the uncertainty of this relationship. Stripe reports that the data volume associated with its Ledger and Data Quality cosystem grew tenfold, but does not specify the time interval required to annualize this figure [12]. Furthermore, PayPal reported 7% growth in total payment volume in 2025, while its number of payment transactions decreased by 4% [13]. These differences demonstrate that business volume, transaction count, and storage growth should not be treated as interchangeable metrics.

Consequently, this project adopts an annual data growth of 34%, based on Stripe's latest reported year-over-year payment volume growth. This percentage is a planning assumption rather than a measured storage growth rate or a prediction of Stripe's actual data growth. Appliedto the baseline storage estimate established in WL-07, the model projects an increase from 150 TB to approximately 201 TB of raw data after one year, assuming a constant retention window and event-size distribution.

4.1.9 WL-09: Concurrent Users

Stripe does not publicly disclose the number of concurrent users interacting with its global payment-processing platform. This metric cannot be directly inferred from API request rates because requests may originate from merchant servers, automated integrations, background jobs, and other system components, rather than from individual users.

As a reference for peak workload, Stripe reported a maximum rate of 27,395 API requests per second during the 2023 Black Friday and Cyber Monday period [**?**]. However, this figure represents request throughput rather than the number of concurrent users.

To support capacity planning, the number of concurrent in-flight requests can be approximated using Little's Law [**?**]:

$$N = \lambda W$$

where N is the average number of requests in the system, λ is the request arrival rate, and W is the average time spent in the system. Assuming a peak arrival rate of 27,395 requests per second and an average response time of 100 ms, the estimated concurrency is:

$$N = 27{,}395 \times 0.1 \approx 2{,}740$$

The 100 ms value is an illustrative design assumption, not a measured average latency reported by Stripe. Consequently, the estimate represents approximately 2,740 concurrent in-flight requests under the specified scenario, not 2,740 concurrent users.

4.1.10 WL-10: Geographic Distribution of Traffic

Stripe's Ledger engineering article describes a global payments network operating across 185 countries and supporting more than 135 currencies and payment methods [12]. This establishes the geographic scope and international nature of the workload.

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